Figure 4.1 Shows the following details:

- **Detect User Mood using MCQ Survey and ML**:
    - The system detects the user's mood by asking them multiple-choice questions (MCQs) in a survey.
    - Machine Learning (ML) algorithms process the survey responses to analyse the user's body features, mood, occasion, preferences and so on.
- **User Inputs Preferences & Mood Data**:
    - The user provides input about their clothing preferences and the mood data derived from the ML analysis is incorporated.
- **Recommend Outfits Based on Mood and Preferences (NLP)**:
    - Using Natural Language Processing (NLP), the system recommends outfits that match the user's mood and stated preferences.
- **Provide Realistic Virtual Try-On Experience (Computer Vision)**:
    - The user can see how the recommended outfits will look on them through a virtual try-on feature powered by computer vision technology.
- **Interactive Chatbot for Fashion Advice**:
    - An AI-driven chatbot is available to provide additional fashion advice or to assist the user with queries about the recommendations.
- **Learn from User Interactions**:
    - The system uses feedback and data from user interactions to improve its recommendations and become more tailored to individual users over time.
- **Optimize System for Accuracy & Efficiency**:
    - Continuous optimization is carried out to enhance the system's accuracy in recommendations and overall efficiency.
- **Showcase Personalized Recommendations**:
    - Based on all the processed data, the system provides the user with personalized fashion recommendations.

4.10 Use Case Model

1. The project will successfully demonstrate the integration of ML for mood detection, computer vision for virtual try-on, and NLP for chatbot interaction, creating a personalized and engaging fashion experience on Android devices.
2. The system should achieve high accuracy in mood classification and outfit recommendations, ensuring relevant suggestions aligned with user preference.

3. The virtual try-on feature should provide a realistic and immersive experience, enhancing the user's ability to visualize outfits before making a decision.
4. The interactive chatbot shall offer personalized fashion advice, improving user engagement.
5. The app should effectively learn from user interactions, providing increasingly accurate outfit suggestions over time. The system’s performance shall be optimized to achieve an efficiency of 80% or higher, ensuring high-quality results for users.
6. The integration of on-device processing shall allow for seamless real-time functionality, providing a smooth and convenient user experience.
7. Through this project, the potential of AI and computer vision for personalized fashion applications on mobile platforms shall be effectively showcased.